

Kunal Maheshwari

Mount Pleasant, MI, USA | (989) 906-2108 | kunal.maheshwari.work@gmail.com | <https://www.linkedin.com/in/kunalmaheshwari26>

SUMMARY

OPT-EAD holder with full-time work authorization and over 4 years of experience as a Software and Data Engineer, skilled in building scalable microservices and distributed data pipelines across diverse domains. Expertise includes Java, Python, Spark, SQL, AWS, and Azure, coupled with CI/CD automation and advanced work on large-scale data platforms and privacy-preserving AI. Seeking to leverage a strong foundation in distributed systems and web technologies to excel as a Staff Software Engineer.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Engineered high-performance REST and SOAP microservices using Java 8/11 and Spring Boot, achieving 99.99 percent SLA uptime for over 50,000 daily API requests.
- Automated CI/CD pipelines with Jenkins and Azure DevOps, enabling continuous delivery across Azure Cloud and Oracle WebLogic environments.
- Modernized legacy systems by migrating to Log4j 2.x, reducing issue resolution time by 20 percent through enhanced monitoring and error handling.
- Spearheaded an EDX platform upgrade, streamlining application architecture and dependency management to deliver features 50 percent faster.
- Collaborated in Agile/Scrum teams to deliver FHIR-compliant healthcare solutions, utilizing Git, SVN, and JUnit for robust testing.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Developed enterprise applications and RESTful web services using Java and EJB for automotive clients, improving overall system performance by 90 percent through SQL query optimization and legacy code modernization.
- Implemented CI/CD pipelines leveraging Jenkins and GitLab, integrating automated testing and quality gates to streamline development workflows.
- Mentored junior developers by conducting code reviews, facilitating debugging sessions, and promoting adherence to clean coding practices.
- Contributed to the development of distributed systems, ensuring robust and scalable solutions for complex client requirements.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems with over 99 percent uptime through proactive monitoring and performance optimization.
- Enhanced SQL data pipelines and database operations, improving query performance by 10 percent and ensuring data integrity.
- Implemented comprehensive unit testing using JUnit4, achieving 99 percent code coverage for critical application modules.
- Supported release planning and sprint execution using JIRA within Agile development teams, facilitating efficient project workflows.

PROJECTS

NYU Taxi Data Analysis Platform (Spark, Python, AWS, Terraform) January 2024 - May 2025

- Developed a distributed Spark-based data pipeline to process over 8 GB of NYC taxi trip data in near real-time, leveraging PySpark for efficient data manipulation.
- Engineered an interactive Flask web application for data exploration, featuring dynamic filtering and visualization capabilities using modern web technologies.
- Containerized ingestion, processing, and analytics components with Docker for reproducible and scalable deployments across cloud environments.
- Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40 percent for large datasets.
- Automated cloud infrastructure provisioning using Terraform across AWS services including S3, RDS, ECS, Lambda, and QuickSight for seamless deployment.

Privacy-Preserving AI Email Client (Capstone Project) (Hugging Face, T5, LoRA) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models using Hugging Face libraries to perform email summarization and classification.
- Applied parameter-efficient fine-tuning with LoRA, supporting low-resource training under stringent privacy constraints for sensitive data.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure and efficient local deployment of AI models.

SKILLS

- Programming Languages: Java, Python, SQL.
- Frameworks and Libraries: Spring Boot, Spring Framework, Apache Spark, PySpark, Hugging Face, T5, LoRA, JUnit, Flask.
- Web Services and Technologies: REST API, SOAP API, gRPC.
- Cloud Platforms: AWS (EC2, RDS, S3, Lambda), Azure (Functions, DevOps, ServiceBus), Google Cloud Platform.

- Databases: PostgreSQL, Oracle Database, IBM DB2, RDS.
- Tools and Technologies: Docker, Kubernetes, Jenkins, Git, SVN, Maven, JIRA, Confluence, Tableau, DBeaver, Terraform.
- Concepts: ETL Pipelines, Data Warehousing, Data Analytics, Agile, Scrum, Test Driven Development, Object Oriented Programming, CI/ CD, Microservices, Distributed Systems, Large-scale data platforms, Infrastructure automation, Privacy-preserving LLM-based applications, Product Management principles.
- Data Analysis and Visualization: Tableau, QuickSight.